A SHORT EXACT SEQUENCE OF MODULES PRODUCES A LONG EXACT SEQUENCE OF EXT GROUPS

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Allowed standard prerequisites (editorial).

- Free resolutions exist and Hom into a free module is exact. Used at the first two lines of the counted proof (lines 17862-17864). Makes the Hom-complex of a resolution a short exact sequence of complexes, since the resolution terms are free.
- The snake lemma for a short exact sequence of complexes. Invoked at the last line of the counted proof (lines 17872-17873). Produces the connecting maps and the long exact sequence in cohomology from that short exact sequence of complexes.

Lemma 0.1. *Let R be a ring. Let M be an R-module. Let $0 \to N' \to N \to N'' \to 0$ be a short exact sequence. Then we get a long exact sequence*

$$\begin{aligned} 0 \to \mathrm{Hom}_R(M, N') \to \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(M, N'') \\ \to \mathrm{Ext}^1_R(M, N') \to \mathrm{Ext}^1_R(M, N) \to \mathrm{Ext}^1_R(M, N'') \to \ldots \end{aligned}$$

Proof. Pick a free resolution $F_\bullet \to M$. Since each of the F_i are free we see that we get a short exact sequence of complexes

$$0 \to \mathrm{Hom}_R(F_\bullet, N') \to \mathrm{Hom}_R(F_\bullet, N) \to \mathrm{Hom}_R(F_\bullet, N'') \to 0$$

Thus we get the long exact sequence from the snake lemma applied to this. □

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